ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $49x^2 + 56xy + 16y^2$	Use the square identity.
2	**(c)** $6x^2 + 19x + 15$	Multiply term by term.
3	**(a)** 45	$a^2 + b^2 = (a-b)^2 + 2ab$.
4	**(b)** 7	Square both sides.
5	**(b)** 1006009	Write $1003 = 1000 + 3$
6	**(b)** 5600	Use $a^2 - b^2 = (a+b)(a-b)$.
7	**(a)** $(x-7)(x+4)$	Find the pair of numbers.
8	**(d)** 32	$4xy = (x+y)^2 - (x-y)^2$.
9	**(b)** $36x^2 - 60xy + 25y^2$	Use the square identity.
10	**(c)** $6x^2 + 25x + 14$	Multiply term by term.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Multiply term by term (1); $= 6x^2 + 17x + 5$ (1).	1+1
12	$x^2 + y^2 = (x+y)^2 - 2xy$ (1); $= 121 - 56 = 65$ (1).	1+1
13	$1002 = 1000 + 2$ (1); $1002^2 = 1004004$ (1).	1+1
14	Multiply term by term (1); $= 6x^2 - x - 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2 + \frac{1}{x^2} = (6)^2 - 2 = 34$ (1); $\left(x - \frac{1}{x}\right)^2 = 34 - 2 = 32$ (2).	1+2
16	$a^2 + b^2 = 4^2 + 2(21)$ (1); $= 58$ (1); $(a+b)^2 = 16 + 4(21) = 100$ (1).	1+1+1
17	Split the middle term: $11x = 8x + 3x$ (1); Group the terms (1); $= (2x+3)(x+4)$ (1).	1+1+1
	OR: $x^2 + \frac{1}{x^2} = (8)^2 - 2 = 62$ (1); $\left(x - \frac{1}{x}\right)^2 = 62 - 2 = 60$ (2).	1+2
18	$a^2 + b^2 = 6^2 + 2(40)$ (1); $= 116$ (1); $(a+b)^2 = 36 + 4(40) = 196$ (1).	1+1+1
	OR: Split the middle term: $16x = 15x + x$ (1); Group the terms (1); $= (3x+1)(x+5)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 - y^2 = (x+y)(x-y)$ (1); $= 12 \times 4 = 48$ (1); $4xy = (x+y)^2 - (x-y)^2 = 144 - 16$, so $xy = 32$ (1); $2(x^2 + y^2) = (x+y)^2 + (x-y)^2 = 160$ (1); $x^2 + y^2 = 80$ (1).	1+1+1+1+1
	OR: $(a+b)^2 = a^2 + b^2 + 2ab$ (1); $81 = a^2 + b^2 + 28$ (1); $a^2 + b^2 = 53$ (1); $(a-b)^2 = (a+b)^2 - 4ab = 81 - 56 = 25$ (2).	1+1+1+2
20	$x^2 - y^2 = (x+y)(x-y)$ (1); $= 10 \times 6 = 60$ (1); $4xy = (x+y)^2 - (x-y)^2 = 100 - 36$, so $xy = 16$ (1); $2(x^2 + y^2) = (x+y)^2 + (x-y)^2 = 136$ (1); $x^2 + y^2 = 68$ (1).	1+1+1+1+1
	OR: $(a+b)^2 = a^2 + b^2 + 2ab$ (1); $144 = a^2 + b^2 + 70$ (1); $a^2 + b^2 = 74$ (1); $(a-b)^2 = (a+b)^2 - 4ab = 144 - 140 = 4$ (2).	1+1+1+2

Mistakes to avoid:

- Writing $(a+b)^2 = a^2 + b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2 = a^2 - b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3 - 3a^2b + 3ab^2 - b^3$.
- Not squaring the whole term: $(2x+3)^2 = 4x^2 + 12x + 9$, not $2x^2 + 12x + 9$.
- Factorising $x^2 + 5x + 6$ with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x-3)(x+3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 73	$a^2+b^2=(a-b)^2+2ab$.
2	**(b)** 14	Square both sides.
3	**(b)** 994009	Write $997=1000-3$
4	**(d)** 5400	Use $a^2-b^2=(a+b)(a-b)$.
5	**(b)** $(x-6)(x+7)$	Find the pair of numbers.
6	**(d)** 40	$4xy=(x+y)^2-(x-y)^2$.
7	**(a)** $81x^2+36xy+4y^2$	Use the square identity.
8	**(c)** $10x^2-7x-12$	Multiply term by term.
9	**(b)** 74	$a^2+b^2=(a-b)^2+2ab$.
10	**(c)** 23	Square both sides.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2+y^2=(x+y)^2-2xy$ (1); $=169-72=97$ (1).	1+1
12	$998=1000-2$ (1); $998^2=996004$ (1).	1+1
13	Multiply term by term (1); $=8x^2+18x-5$ (1).	1+1
14	$x^2+y^2=(x+y)^2-2xy$ (1); $=225-108=117$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+\frac{1}{x^2}=(9)^2-2=79$ (1); $\left(x-\frac{1}{x}\right)^2=79-2=77$ (2).	1+2
16	$a^2+b^2=3^2+2(28)$ (1); $=65$ (1); $(a+b)^2=9+4(28)=121$ (1).	1+1+1
17	Split the middle term: $7x=4x+3x$ (1); Group the terms (1); $=(2x+1)(3x+2)$ (1).	1+1+1
	OR: $x^2+\frac{1}{x^2}=(10)^2-2=98$ (1); $\left(x-\frac{1}{x}\right)^2=98-2=96$ (2).	1+2
18	$a^2+b^2=8^2+2(33)$ (1); $=130$ (1); $(a+b)^2=64+4(33)=196$ (1).	1+1+1
	OR: Split the middle term: $11x=8x+3x$ (1); Group the terms (1); $=(4x+3)(x+2)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2-y^2=(x+y)(x-y)$ (1); $=14\times 8=112$ (1); $4xy=(x+y)^2-(x-y)^2=196-64$, so $xy=33$ (1); $2(x^2+y^2)=(x+y)^2+(x-y)^2=260$ (1); $x^2+y^2=130$ (1).	1+1+1+1+1
	OR: $(a+b)^2=a^2+b^2+2ab$ (1); $100=a^2+b^2+48$ (1); $a^2+b^2=52$ (1); $(a-b)^2=(a+b)^2-4ab=100-96=4$ (2).	1+1+1+2
20	$x^2-y^2=(x+y)(x-y)$ (1); $=16\times 4=64$ (1); $4xy=(x+y)^2-(x-y)^2=256-16$, so $xy=60$ (1); $2(x^2+y^2)=(x+y)^2+(x-y)^2=272$ (1); $x^2+y^2=136$ (1).	1+1+1+1+1
	OR: $(a+b)^2=a^2+b^2+2ab$ (1); $225=a^2+b^2+112$ (1); $a^2+b^2=113$ (1); $(a-b)^2=(a+b)^2-4ab=225-224=1$ (2).	1+1+1+2

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2=a^2-b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3-3a^2b+3ab^2-b^3$.
- Not squaring the whole term: $(2x+3)^2=4x^2+12x+9$, not $2x^2+12x+9$.
- Factorising x^2+5x+6 with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2-18=2(x-3)(x+3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1010025	Write $1005 = 1000 + 5$
2	**(c)** 10400	Use $a^2 - b^2 = (a+b)(a-b)$.
3	**(c)** $(x-8)(x-6)$	Find the pair of numbers.
4	**(a)** 84	$4xy = (x+y)^2 - (x-y)^2$.
5	**(a)** $64x^2 - 48xy + 9y^2$	Use the square identity.
6	**(c)** $12x^2 - 5x - 2$	Multiply term by term.
7	**(b)** 109	$a^2 + b^2 = (a-b)^2 + 2ab$.
8	**(c)** 47	Square both sides.
9	**(a)** 990025	Write $995 = 1000 - 5$
10	**(c)** 4000	Use $a^2 - b^2 = (a+b)(a-b)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$1004 = 1000 + 4$ (1); $1004^2 = 1008016$ (1).	1+1
12	Multiply term by term (1); $= 15x^2 - 14x - 8$ (1).	1+1
13	$x^2 + y^2 = (x+y)^2 - 2xy$ (1); $= 81 - 40 = 41$ (1).	1+1
14	$996 = 1000 - 4$ (1); $996^2 = 992016$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2 + \frac{1}{x^2} = (11)^2 - 2 = 119$ (1); $\left(x - \frac{1}{x}\right)^2 = 119 - 2 = 117$ (2).	1+2
16	$a^2 + b^2 = 5^2 + 2(36)$ (1); $= 97$ (1); $(a+b)^2 = 25 + 4(36) = 169$ (1).	1+1+1
17	Split the middle term: $19x = 15x + 4x$ (1); Group the terms (1); $= (3x+2)(2x+5)$ (1).	1+1+1
	OR: $x^2 + \frac{1}{x^2} = (12)^2 - 2 = 142$ (1); $\left(x - \frac{1}{x}\right)^2 = 142 - 2 = 140$ (2).	1+2
18	$a^2 + b^2 = 9^2 + 2(52)$ (1); $= 185$ (1); $(a+b)^2 = 81 + 4(52) = 289$ (1).	1+1+1
	OR: Split the middle term: $17x = 15x + 2x$ (1); Group the terms (1); $= (5x+1)(2x+3)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 - y^2 = (x+y)(x-y)$ (1); $= 20 \times 6 = 120$ (1); $4xy = (x+y)^2 - (x-y)^2 = 400 - 36$, so $xy = 91$ (1); $2(x^2 + y^2) = (x+y)^2 + (x-y)^2 = 436$ (1); $x^2 + y^2 = 218$ (1).	1+1+1+1+1
	OR: $(a+b)^2 = a^2 + b^2 + 2ab$ (1); $121 = a^2 + b^2 + 56$ (1); $a^2 + b^2 = 65$ (1); $(a-b)^2 = (a+b)^2 - 4ab = 121 - 112 = 9$ (2).	1+1+1+2
20	$x^2 - y^2 = (x+y)(x-y)$ (1); $= 18 \times 10 = 180$ (1); $4xy = (x+y)^2 - (x-y)^2 = 324 - 100$, so $xy = 56$ (1); $2(x^2 + y^2) = (x+y)^2 + (x-y)^2 = 424$ (1); $x^2 + y^2 = 212$ (1).	1+1+1+1+1
	OR: $(a+b)^2 = a^2 + b^2 + 2ab$ (1); $169 = a^2 + b^2 + 72$ (1); $a^2 + b^2 = 97$ (1); $(a-b)^2 = (a+b)^2 - 4ab = 169 - 144 = 25$ (2).	1+1+1+2

Mistakes to avoid:

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- Using $(a-b)^2 = a^2 - b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3 - 3a^2b + 3ab^2 - b^3$.
- Not squaring the whole term: $(2x+3)^2 = 4x^2 + 12x + 9$, not $2x^2 + 12x + 9$.
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